

Towards A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

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Motivation

- Climate change-driven large, destructive wildfires becoming more frequent, especially in California

- Prescribed burns are known to be effective in reducing wildfire intensity in some scenarios, yet:
  - Environmental trends indicate the window for prescribed burns is shrinking
  - Limited implementation due to controversy on risk, cost, and environmental impact

- Much of the existing body of research was performed in the previous century and did for climate change's impact on prescribed burn efficacy

- Statistical analysis can be used to identify effective or ineffective prevention strategies, such as prescribed burns, but the data needed to study this question is fragmented and complex

Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

We Investigate these questions with:

- Monthly Time Series Analysis – Correlation and Cross-Correlation
- Inferential Statistics - Mann-Whitney U-test
- Regression Analysis – Ordinary Least Squares Linear Regression

Data

Variable	Source	Description	Range	Units
Acres Burned	CALfire	Wildfire acres burned	0+	Acres
Treatment Acres	CALfire	Prescribed fire acres burned	0+	Acres
Soil Moisture	NOAA	Soil moisture levels calculated based on temperature and precipitation	-500 to 500	mm
Precipitation	NOAA	Monthly precipitation averaged across the state	0+	inches
Temperature	NOAA	Average, minimum, and maximum temperature averaged across the state	0+	Farenhiet
Palmer Index	NOAA	Drought Severity index, hydrological drought index, modified drought index, Z-index	-10 to 10	NA

Table 1: Variables in the dataset, including descriptions, sources, and units

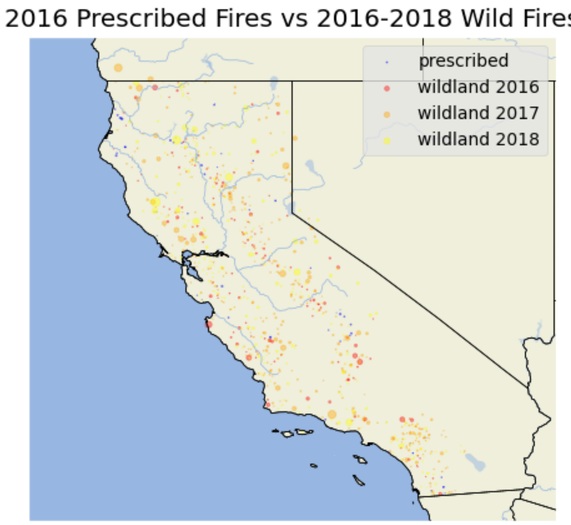
Dataset Creation

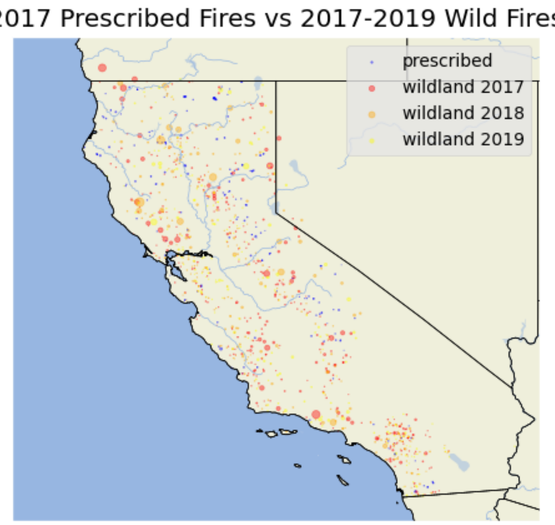
- Data from 1734 wildfires from 1/2016 – 12/2022
- Columns:
  - wildfire acres burned, previous month precipitation, avg temp, max temp, min temp, drought index, soil moisture
  - for each 3/6/12/24/36 month time frame, 5/10/25/50/100 km distance, total prescribed fire acres burned
- Monthly dataset: sums total acres per month



Figure 1: Distribution of wildfire size

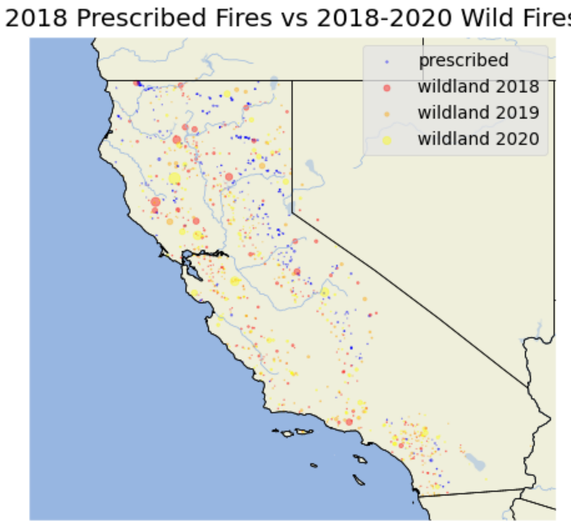
Data Overview











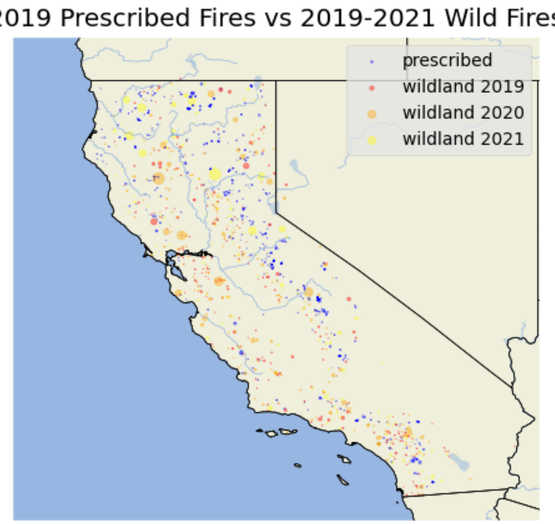




Figure 2: Visualization of prescribed fires for each year and wildfires in following years

Time Series Analysis: Monthly

Wildfire acres burned have the strongest correlation with climate variables

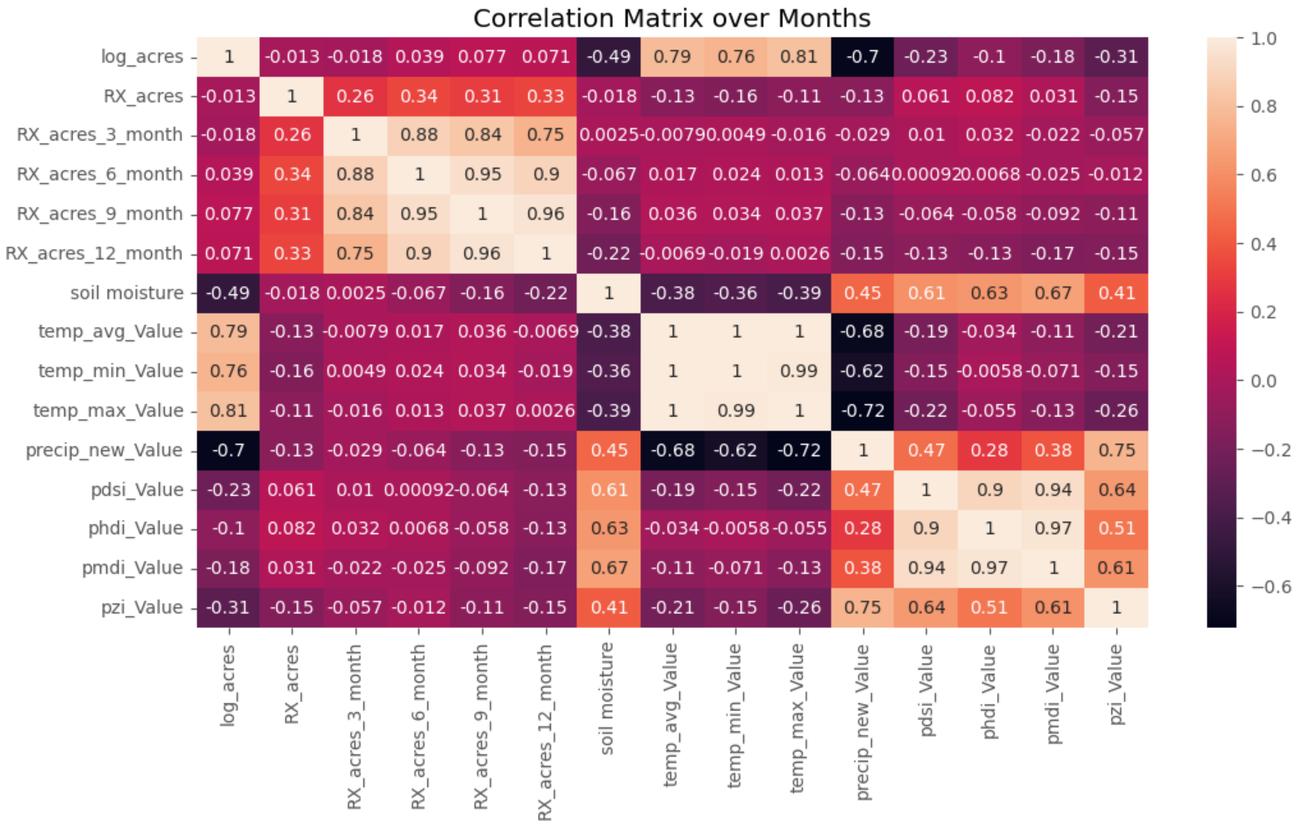


Figure 3: Correlation matrix for predictors and wildfire size

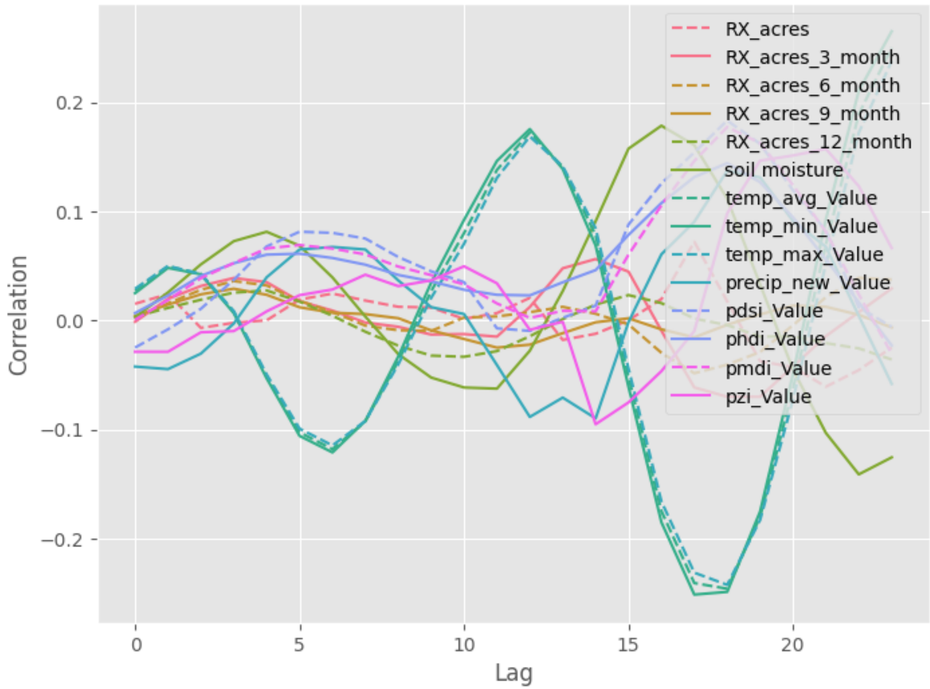


Figure 4: Cross-correlations for predictors and wildfire size

Inferential Analysis: Treatment Data

Figure 5: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

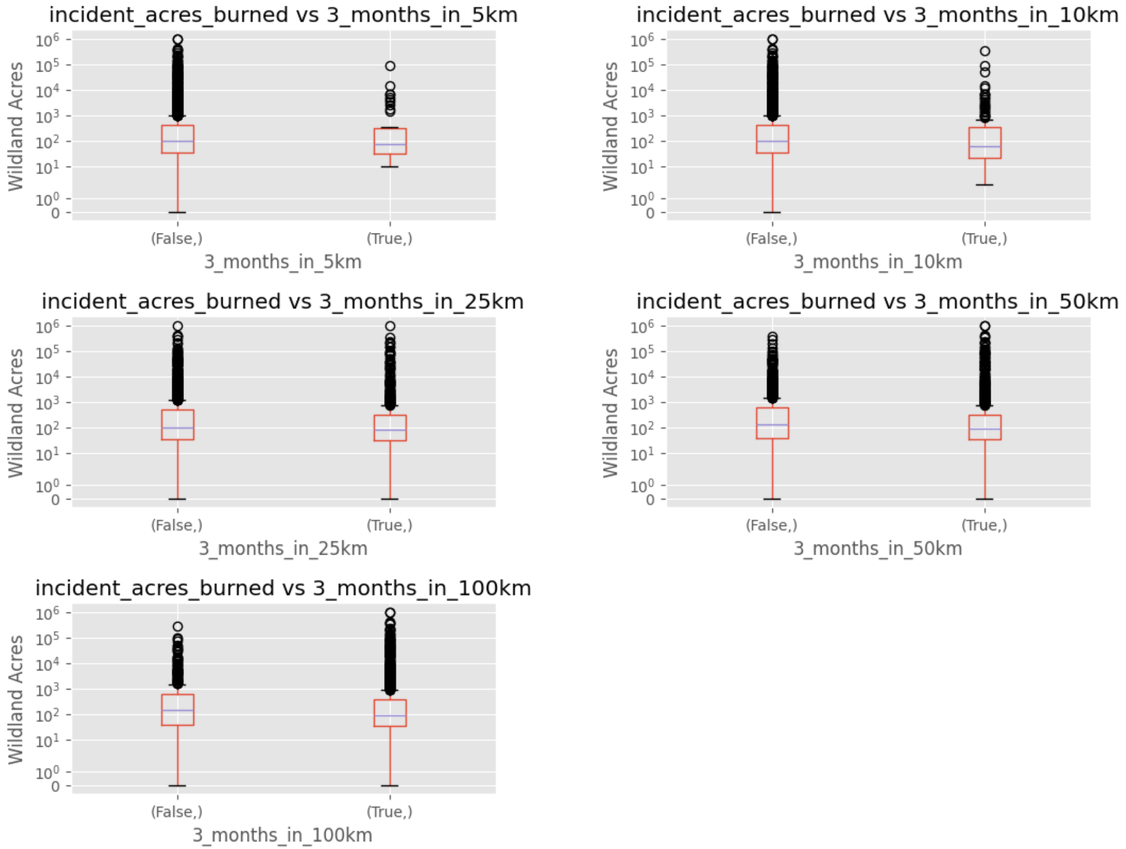


Table 2: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
3_months_in_5km	MWU	34773.5	two-sided	0.596817	0.046675	0.476663
3_months_in_10km	MWU	89038.5	two-sided	0.010317	0.134713	0.432644
3_months_in_25km	MWU	288042.0	two-sided	0.016372	0.072567	0.463716
3_months_in_50km	MWU	318792.0	two-sided	0.001780	0.088990	0.455505
3_months_in_100km	MWU	200542.0	two-sided	0.044303	0.072855	0.463573

Prescribed burn treatment is associated with fewer acres burned at 10 km, 25 km, 50 km, and 100km distance when completed 3 months before a fire

Inferential Analysis: Treatment Data

Figure 6: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

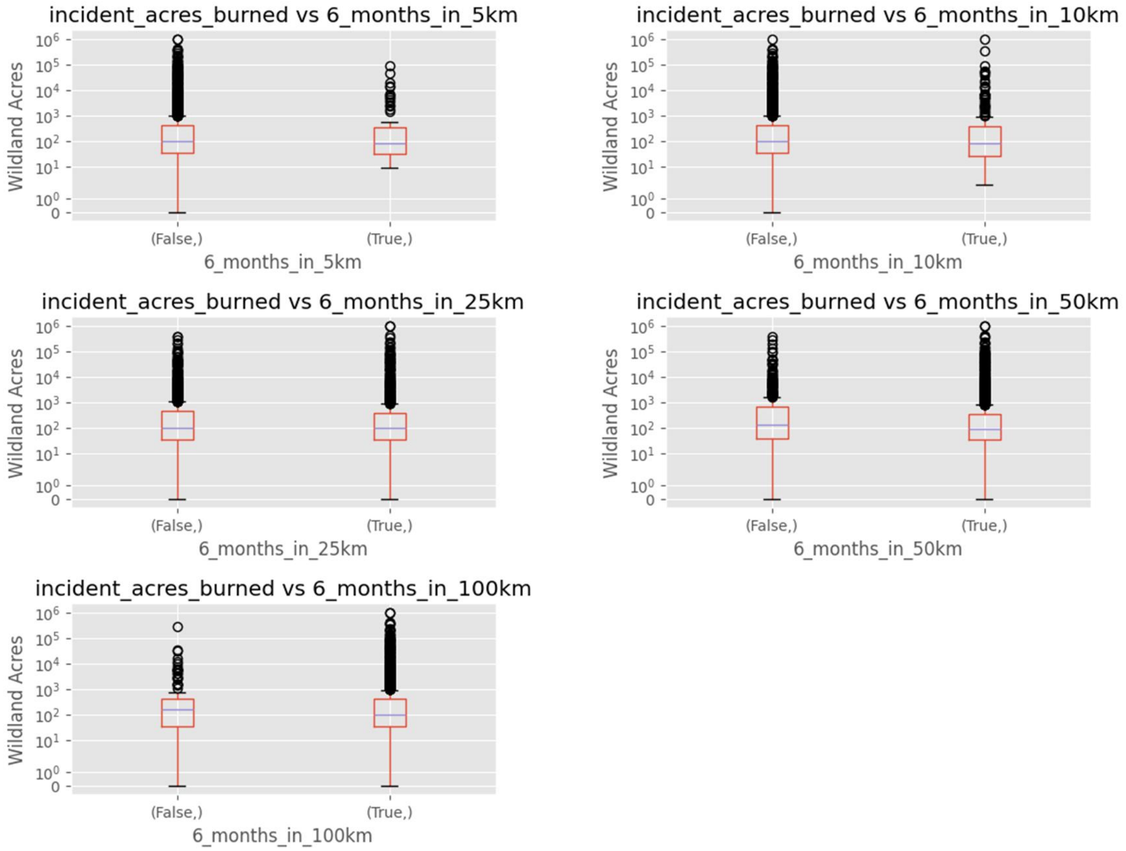


Table 3: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
6_months_in_5km	MWU	53644.5	two-sided	0.930174	0.006363	0.496819
6_months_in_10km	MWU	144932.5	two-sided	0.052046	0.082357	0.458821
6_months_in_25km	MWU	348994.0	two-sided	0.285439	0.030002	0.484999
6_months_in_50km	MWU	238604.0	two-sided	0.008402	0.086858	0.456571
6_months_in_100km	MWU	88006.5	two-sided	0.231858	0.065634	0.467183

Prescribed burn treatment is associated with fewer acres burned at 10 km and 50 km distance when completed 6 months before a fire

Regression Analysis: Treatment

$\hat{Y} = 5.322 - 1.232x_{PZI} - 3.005x_{RX}$

Y: Wildfire acres burned

X: PZI drought index, 3 months prescribed fire in 10km (RX)

- $R^2 = 0.827$
- Train $MSE = 5.059$
- $F = 2771, \mathrm{p} < 0.001$

Prescribed burns are related to decreases in wildfire size

Increased drought severity is related to increases in wildfire size

Prescribed burns have a slightly greater effect on wildfire size than drought

	coef	std err	t	P>|t|	[0.025	0.975]
pzi_Value	-1.2317	0.424	-2.907	0.004	-2.063	-0.401
prescribed_3_months_size_10km	-3.0052	1.429	-2.102	0.036	-5.809	-0.202
constant	5.3221	0.145	36.626	0.000	5.037	5.607

Table 4: Regression coefficient results

Regression Analysis: Treatment

Figure 7: Residuals and partial regression for PZI

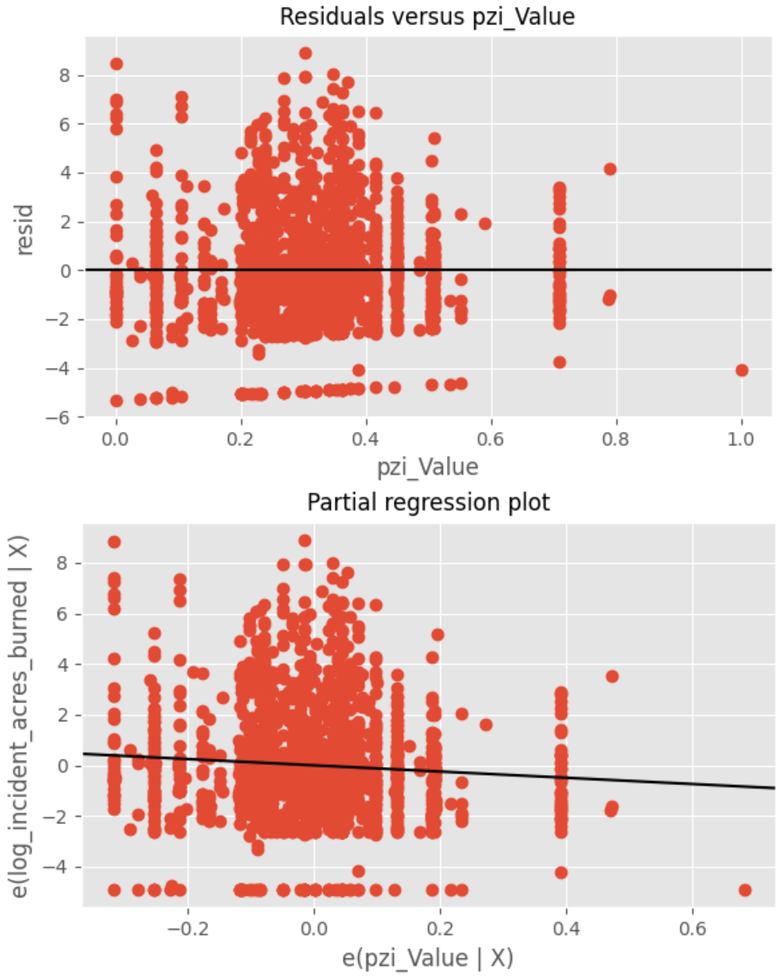
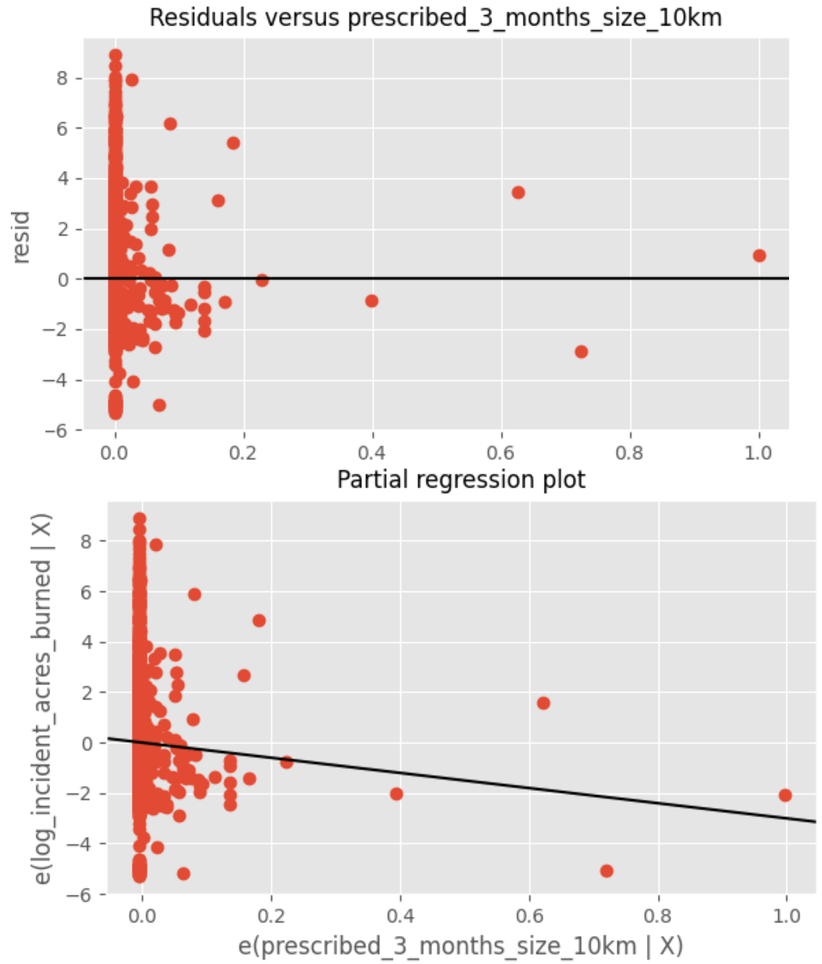


Figure 8: Residuals and partial regression for prescribed burns



Regression Analysis: Drought Index

$$Y = -9.813 + 0.097x_t + 1.226X_p$$

Y: PZI drought index

X: maximum temperature, precipitation

Precipitation decreases are related to increased drought severity

Precipitation has the greatest impact on drought index

Temperature has very minimal effect on drought index

- $R^2 = 0.737$
- Train $MSE = 1.303$
- $F = 113.3, \mathrm{p} < 0.001$

	coef	std err	t	P>|t|	[0.025	0.975]
temp_max_Value	0.0973	0.013	7.280	0.000	0.071	0.124
precip_new_Value	1.2261	0.085	14.374	0.000	1.056	1.396
intercept	-9.8125	1.090	-9.004	0.000	-11.981	-7.644

Table 5: Regression coefficient results

Regression Analysis: Drought Index

Figure 9: Residuals and partial regression for temperature

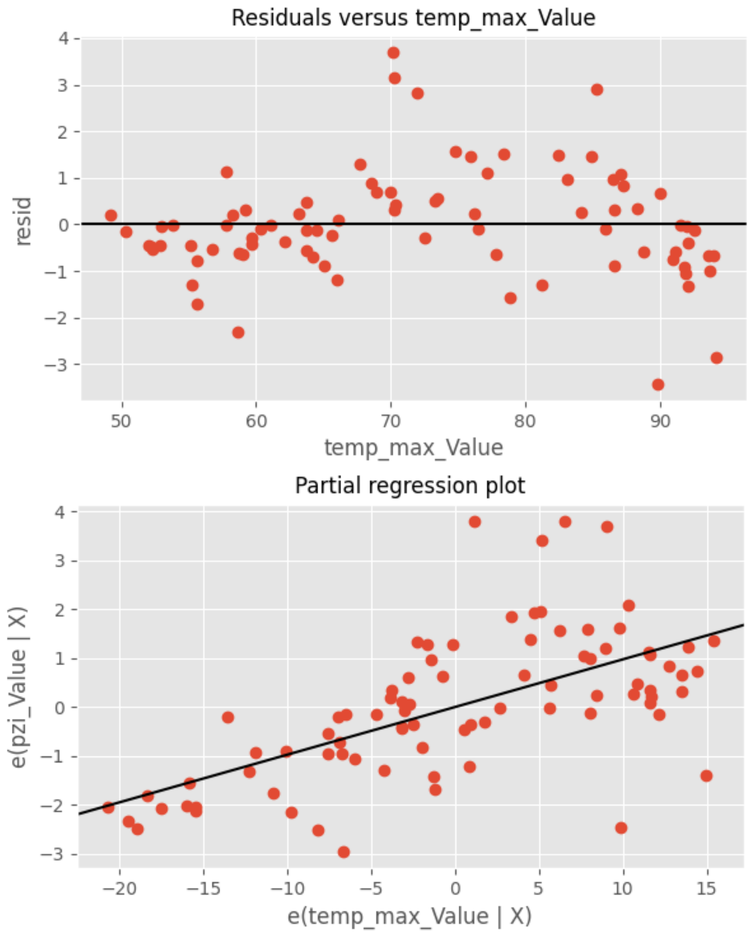
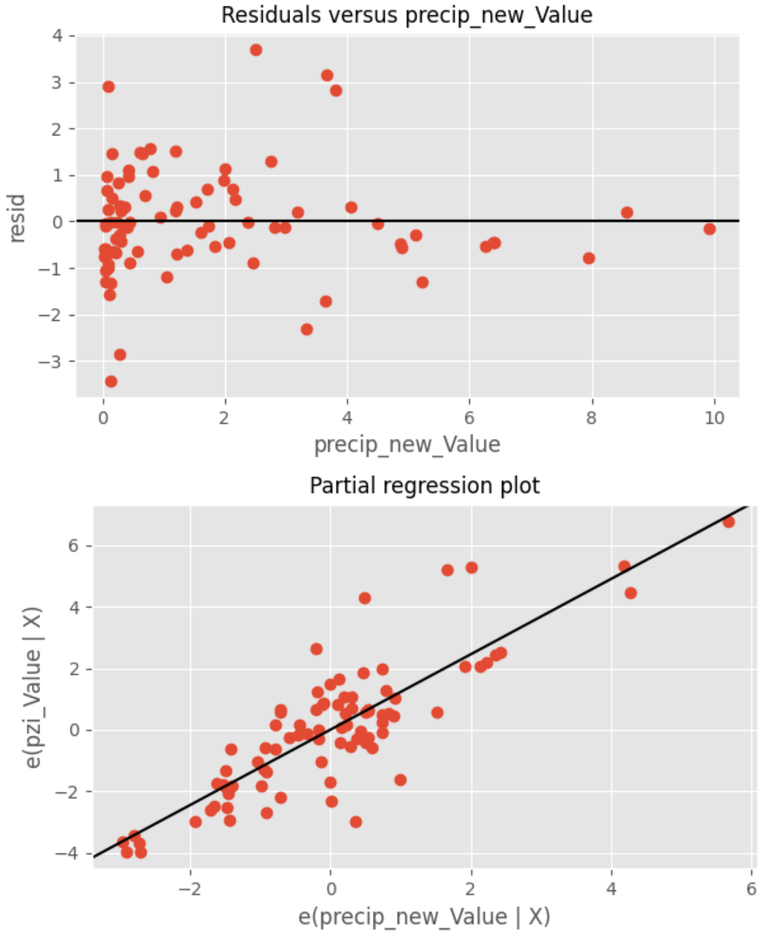


Figure 10: Residuals and partial regression for precipitation



Discussion

- Prescribed burns are associated with decreased wildfire size and have a greater effect than drought index
- When chaining the models, precipitation has the largest impact on wildfire size

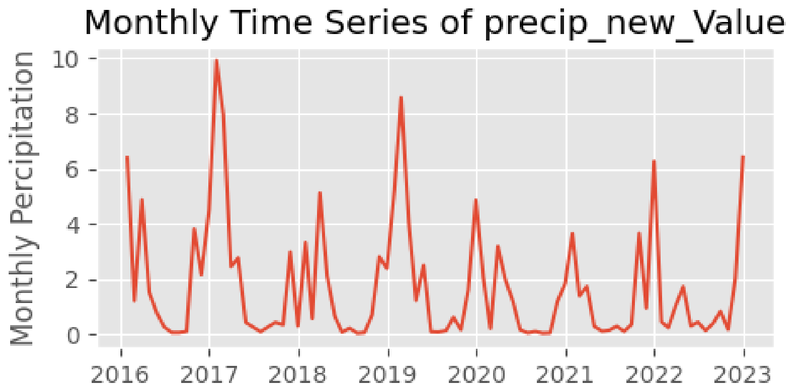


Figure 11: Precipitation time series

Limitations:

- Treatment climate variables are for the whole of California, instead of the region the fire occurs in
- Vegetation type is not considered in this model
- Models use binning for handing spatial and temporal differences – an integrated model could be better

Recall: Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

Conclusion and Future Work

- Increased prescribed burn acreage is related to decreased wildfire size under certain circumstances
- When controlling for drought index, prescribed burns are still a significant predictor of wildfire size
- Prescribed burns have the greatest impact when performed within 3 months and 10 km of wildfires.

- Future analysis will explore other methods for incorporating spatio-temporal information, such as coregionalization modeling and mixed effects modeling



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Methods

- Dataset creation:
  - Monthly Dataset
  - Treatment Dataset
- Inferential analysis:
  - Wilcoxson Signed Rank test / Mann-Whitney U-test
- Time series Analysis
  - Cross correlation
  - Correlation Matrix
- Regression Analysis
  - PZI drought index
  - Acres burned per wildfire incident